SKULPT Yourself: A Data-Driven Facial Reconstruction Pipeline and Expert-Guided Evaluation Study

AEL AI Experience Lab

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TL;DR

Problem While recent studies have *automated* craniofacial reconstruction, *evaluation* heavily depends on geometric error or similarity metrics

Solution *Large-scale, expert-grounded* study of reconstructions with SKULPT and *correlation analysis* with standard similarity methods

Results Evaluations indicate that standard similarity metrics *do not reliably* match expert judgement, implying the need for experts in the loop

TASK

★ Automate **forensic facial reconstruction** from skull images

★ **Effectively evaluate** reconstruction results, while offering insights into what constitutes a coherent reconstruction in forensic contexts

EXISTING METHODS

★ **Previous studies** used statistical modeling, template deformation, superimposition, deep learning to address **reconstruction** task

★ Traditionally, **evaluation** has relied on geometric error metrics, statistical similarity measures, or general surveys involving non-expert observers

DATA

★ 760 skull-face pairs, mixed **CBCT** and **PMCT** scans, from the NFS

★ **Demographic diversity** age, gender, and scan type

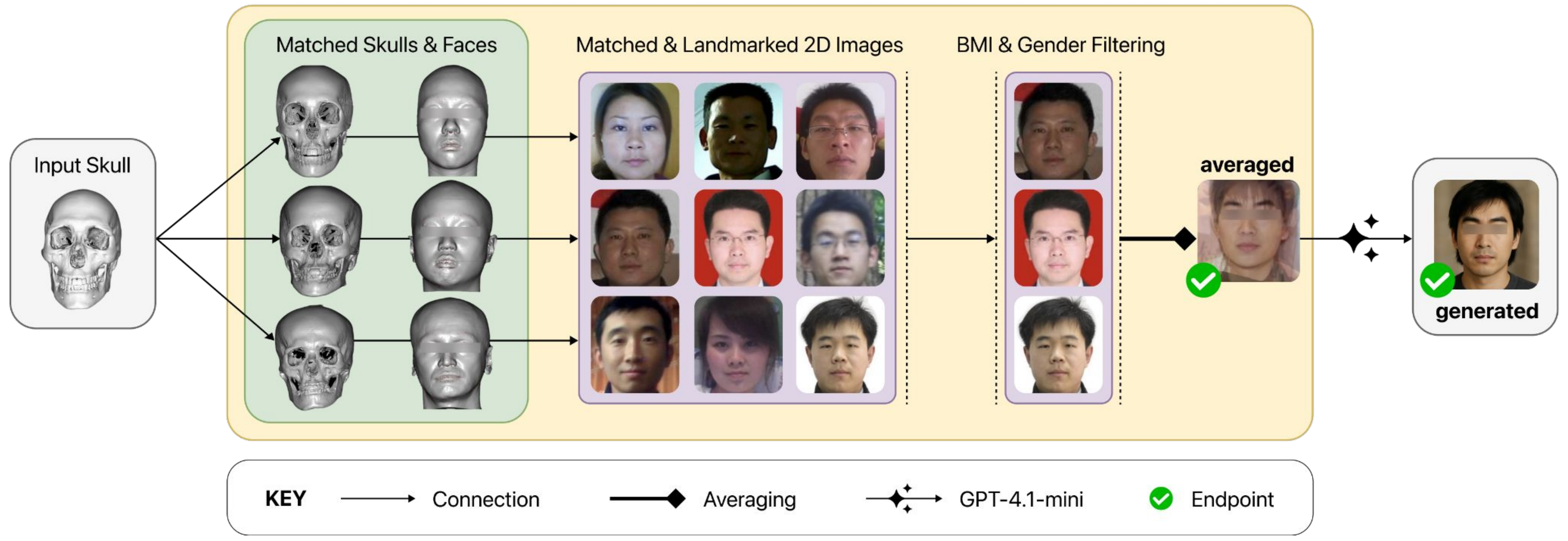
★ **Preprocessing** landmark normalisation, handling missing values, and alignment

PIPELINE CONSTRUCTION

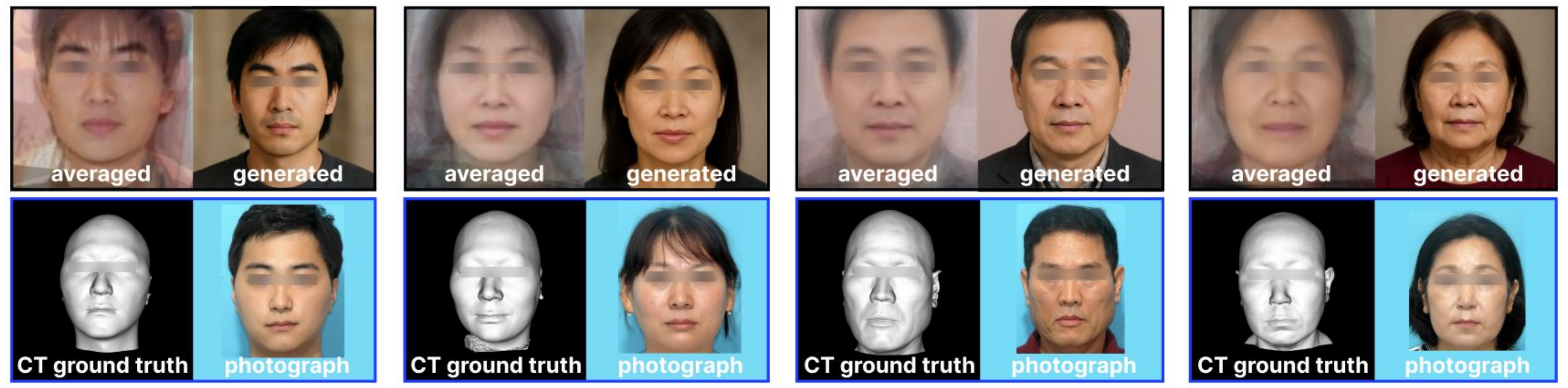
★ **RL agent** courtesy of the NIMS, predicts 20 skull landmarks via a multistage deep reinforcement learning approach

★ Two types of reconstructions:

- **Averaged** Following Euclidean distance-based matching between skull landmarks, averaging via Procrustes distance-based matching and BMI & gender filtering
- **Generated** prompt-based facial generation using GPT-4.1-mini on top of averaged reconstruction



PIPELINE RESULTS



EXPERT SCORING OVERVIEW

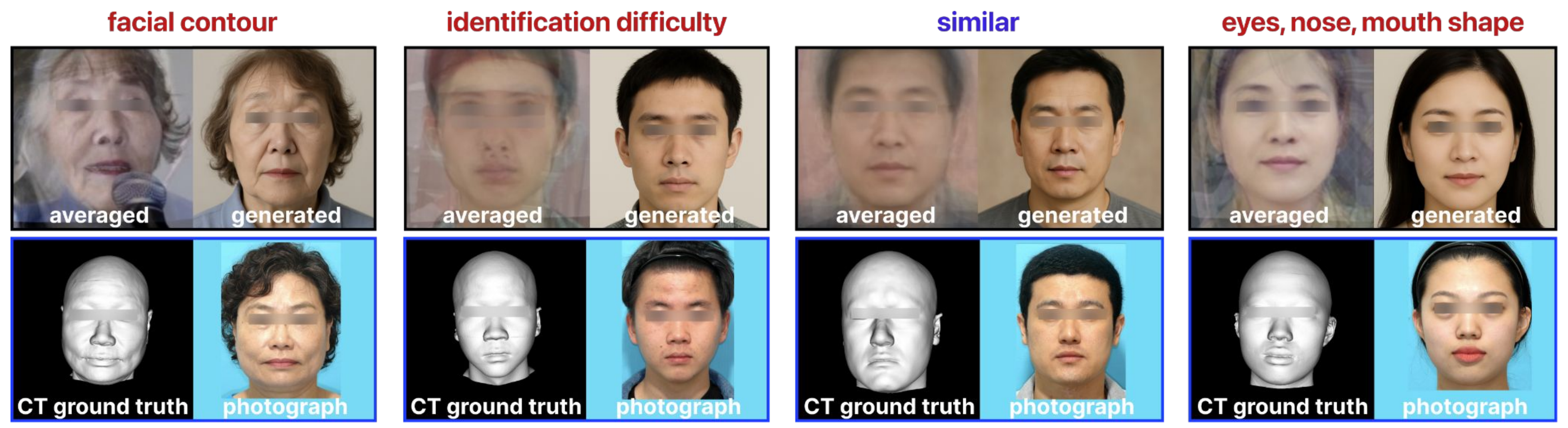
★ Faces scored 1 **(low similarity)** to 7 (high similarity)

★ **Averaged** scored consistently, mostly 3 to 5; **generated** more varied, lower → significantly different distributions (t-test ***p* < 0.03**)

TOPIC-BASED COMMENT ANALYSIS

★ **295** comments, categorised into **16** topic groups (positive vs negative)

★ **Top topics**: face contour, identification difficulty, ethnicity mismatch, feature shape, CT rotation, similar (only for high-score cases)



SIMILARITY METRICS VS EXPERT SCORES

★ **Three** SOTA similarity metrics vs expert scores

★ Correlation **weak** and **inconsistent**

Metric	**ρ (Avg)**	***p*-value (Avg)**	**ρ (Gen)**	***p*-value (Gen)**
LPIPS	**–0.115**	**0.00257**	**–0.121**	**0.0014**
DeepFace (Dlib)	0.007	0.859	**0.113**	**0.00291**
DeepFace (Facenet512)	**0.108**	**0.00443**	**0.195**	**<0.0001**
DeepFace (VGG16)	0.074	0.0532	–0.021	0.588
Face Recognition	**0.091**	**0.0165**	0.004	0.920

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